

SliceMatic

Business Analysis & Unit Economics — A Deep-Dive Reference Model

Rs.511

CONTRIBUTION / ORDER

~13/day

TRUE BREAK-EVEN

~18 mo

CAPITAL PAYBACK

STAGE 1 · PART B

Outlet: Single-unit pizza delivery · New Ashok Nagar, East Delhi

Scope: Cost structure · unit economics · break-even · scenarios · channel & customer economics · risks

Basis: Reference model challenged & corrected for internal consistency

Prepared as: Shareable Stage-1 business analysis (Part B)

00 Executive Summary

SliceMatic is a viable single-outlet QSR with genuinely strong unit economics — but the reference model oversells it. After correcting one structural error, the business is healthy, not spectacular, and its real risks sit in **demand ramp-up** and **aggregator commission**, not in rent or COGS.

What the reference model claims	What the corrected analysis shows
Contribution margin 76% (Rs.644/order)	Contribution margin ~60% (Rs.511/order) — the model charged one pizza's cost against a ~2-pizza order
Break-even 11 orders/day	Break-even ~13 orders/day own-channel; ~15/day once aggregators are in the mix
Operating margin 58.9% at launch	~43% at mature 60% capacity (own-channel); ~35% blended — and far lower during the 6-month ramp
Rent sensitivity is a headline risk	Rent is not the binding constraint — demand and channel mix are

THE ONE-LINE BUSINESS CASE FOR THE SOFTWARE

Every order placed through the owned digital channel earns **Rs.511** of contribution; the same order via Zomato/Swiggy earns only **Rs.339** after a 25% commission. The ordering system — and its Stage-3 recommendation engine — exists to **convert aggregator-acquired customers into owned, repeat customers**. That Rs.172/order gap, compounded over a customer's lifetime, is the entire ROI story.

01 Market & Competitive Context

SliceMatic launches into the most contested category in Indian QSR. Understanding the battlefield frames every number that follows.

- **Delivery is the format, not a feature.** ~60–70% of Tier-1 pizza volume is delivery/takeaway, and SliceMatic is explicitly delivery-first. The 8 dine-in covers are a walk-in convenience, not a revenue engine, so the unit economics in this document **model delivery orders only** and treat dine-in as immaterial (the conservative choice). Every number that follows therefore stands on delivery throughput and SLA.
- **The aggregator duopoly sets the floor.** Zomato and Swiggy own discovery. A new outlet with no brand will likely see **50–60% of early orders** arrive via aggregator at ~20–25% commission — a structural margin tax until owned demand builds.
- **Incumbents anchor expectations.** Domino's (30-min promise, deep menu, dense store network) and local players have trained customers on speed and discounts. SliceMatic's 30-min guarantee is table stakes, not a differentiator.
- **The moat is data + locality.** A single outlet can't out-spend Domino's. It can win a 4 km radius on freshness, customisation, and a tight repeat-customer loop — which is exactly what the owned ordering system enables.

IMPLICATION FOR THE MODEL

Any unit-economics number quoted "at 60% capacity, own-channel" is a **maturity** figure. The launch reality is lower volume, aggregator-heavy mix, and thinner blended margins. Both views are modelled below.

02 Cost Structure

2.1 Monthly fixed costs — Rs.2,02,910

Incurred regardless of a single order. This is the monthly hurdle the business clears before earning a rupee of profit.

Cost item	Detail	Monthly (Rs.)	% of fixed
Kitchen rent	1,200 sq ft commercial lease	55,000	27.1%
Delivery riders (2)	Fixed salary component only	32,000	15.8%
Head chef	1 experienced pizza chef	28,000	13.8%
Counter + billing	1 person — order & cash	18,000	8.9%
Kitchen helper	Prep staff + ESI/PF	16,500	8.1%
Equipment EMI	Rs.4.5L, 36-month loan	14,500	7.1%
Electricity	Ovens + refrigeration + AC	12,600	6.2%
Marketing (fixed)	Meta retainer + listings	8,000	3.9%
Misc / contingency	3% maintenance buffer	5,760	2.8%
Gas / LPG	3 cylinders/month	5,550	2.7%
Packaging (fixed)	Branded box minimum order	4,200	2.1%
Internet + POS + SaaS	Broadband + POS + gateway fee	2,800	1.4%
TOTAL FIXED COSTS		2,02,910	100%

Observation: **People (riders + chef + helper + counter) = Rs.94,500, or 47% of fixed cost.** Labour, not rent, is the largest controllable block — and the counter role (Rs.18k) is precisely what the digital ordering system makes redundant at the order-taking step.

2.2 Variable costs — per pizza vs per order

The critical distinction the reference model blurs: **some costs are per-pizza, some are per-order.** An order averages ~2 pizzas (AOV Rs.847 ÷ ~Rs.420 selling price/pizza).

Component	Basis	Rate	Per 2-pizza order
Ingredients (dough, sauce, cheese, toppings)	per pizza	Rs.130	Rs.260
Packaging (box + bag)	per pizza	Rs.18	Rs.36
Delivery (rider incentive, 1 trip)	per order	Rs.22	Rs.22
Payment gateway (1.8% of GMV)	per order	1.8%	Rs.18
TOTAL VARIABLE / ORDER			Rs.336

MODELLING TRAP TO AVOID

The reference COGS table embeds the **per-order** delivery cost (Rs.22) inside the **per-pizza** COGS. Taken literally, a 2-pizza order would double-count delivery (Rs.44). Delivery and gateway are charged **once per order**; ingredients and packaging scale per pizza. Getting this split right is what makes the contribution margin defensible.

03 Unit Economics — COGS & Margin by Category

Component	Margherita	Farm House	Cheese Burst	Avg
Base ingredients	38	45	72	52
Sauce + seasoning	12	14	14	13
Cheese (mozzarella)	28	32	65	38
Pizza-specific toppings	8	22	18	17
Add-on topping (1)	10	10	10	10
Packaging	18	18	18	18
Delivery (variable)	22	22	22	22
Total COGS / pizza	136	163	219	170
Selling price (base+pizza+1 topping)	397	417	447	420
Gross margin / pizza	65.7%	60.9%	51.0%	59.5%

Gross margin % by menu category (base / pizza / topping split)

Category	Menu price band	Component cost	Gross margin
Base (crust)	Rs.149 – 229	Rs.38 – 72	~ 68-74%
Pizza (sauce + cheese + toppings)	Rs.299 – 379	~Rs.68	~ 77-82%
Add-on topping	Rs.39 – 69	~Rs.10	~ 74-85%

WHERE THE MARGIN LIVES

Toppings and the pizza layer are the highest-margin components (77-85%). Cheese Burst is the margin laggard (51%) because premium mozzarella loading is expensive. The single highest-ROI revenue lever is therefore **topping attach-rate** — which is exactly what a recommendation engine optimises.

04 Contribution Margin — Reference vs Corrected

This is the single most important correction in the entire model. Contribution margin = revenue minus *all* variable cost; it is what each order contributes toward fixed costs and profit.

Per order (ex-GST)	Reference model	Corrected (~2-pizza)
Revenue (AOV)	847	847
Less: Ingredient COGS	(148)	(260)
Less: Packaging	(18)	(36)
Less: Delivery (per order)	(22)	(22)
Less: Payment gateway	(15)	(18)
Contribution margin	644 (76.0%)	511 (60.3%)

WHY THE REFERENCE IS WRONG BY ~26%

It subtracts **one pizza's** ingredient and packaging cost from a **two-pizza order's** revenue. Add the second pizza's COGS (Rs.130 + Rs.18) and the margin falls from Rs.644 to Rs.511. A 60% contribution margin is still excellent for QSR — but it is the honest number, and it is the one you should defend in the demo.

Reality check: mature single-outlet QSR EBITDA margins typically land at **15-25%** once real marketing/CAC, wastage, aggregator commission, and the demand ramp are loaded in. The model's high margins come from low rent, an owner-operator structure, and a generous capacity assumption. Treat 40%+ operating margins as a best-case ceiling, not a base case.

05 Break-Even & Sensitivity

Scenario	CM / order	Break-even / month	Break-even / day
Reference (overstated)	644	315	~11
Corrected — own channel	511	397	~13
Blended — 40% aggregator	442	459	~15
Blended — 60% aggregator (launch)	408	498	~17
Rent raised to Rs.70k (own channel)	511	426	~14

At the 60%-capacity target of **47 orders/day**, the outlet runs at **~3.5× break-even** on own channel (not the 4.3× the reference claims). It only turns unprofitable below ~13 orders/day — a ~72% collapse from plan. The model is **robust to cost shocks but exposed to demand shortfall**.

RENT IS A LOW-SENSITIVITY LEVER — BUT READ IT CORRECTLY

Per rupee, rent barely moves break-even: each Rs.10,000/month adds only ~0.6 orders/day (Rs. 10,000 ÷ Rs.511 ÷ 30). But the question isn't "how high can rent go *alone*" — fixed costs rarely move one line at a time. A bigger or pricier space usually arrives **bundled** with an extra chef, a third rider, and higher utilities, so fixed cost steps up as a **block**. The honest viability test is therefore **total fixed cost vs contribution**: at planned 60% volume the outlet earns ~Rs.7.2L/month of

contribution, and stays profitable until the whole fixed-cost block approaches that. The real risk is whether each step-up in fixed costs is matched by enough incremental contribution — which is governed by **demand ramp** and **channel mix**, not rent in isolation.

06 Scenario Model — Monthly EBITDA

A single steady-state number hides the truth. Here is the business across four realistic states, each with an explicit volume and channel-mix assumption.

Scenario	Orders/day	Aggregator mix	CM/order	Monthly EBITDA	Op. margin
Worst — launch (M1-2)	20	60%	408	Rs.41,800	~8%
Conservative (M3-4)	30	50%	425	Rs.1,79,600	~24%
Base — mature (M6+)	47	40%	442	Rs.4,20,600	~35%
Optimistic (own-heavy)	57	25%	468	Rs.5,97,400	~41%

THE TWO LEVERS THAT MOVE EBITDA MOST

(1) Volume — moving from 20 to 47 orders/day swings EBITDA ~10×. **(2) Channel mix** — every 10 points shifted from aggregator to owned channel adds ~Rs.10-13/order of contribution. The product roadmap should attack both: frictionless ordering drives volume; the recommendation engine + repeat loop drives owned mix.

07 The Aggregator Decision — Channel Economics

Zomato/Swiggy is the single biggest margin variable in the model. Here is the same order, two ways:

Owned channel	Rs.	Aggregator (25%)	Rs.
Revenue (ex-GST)	847	Revenue (ex-GST)	847
Ingredients + packaging	(296)	Commission (25%)	(212)
Own delivery	(22)	Ingredients + packaging	(296)
Gateway	(18)	Delivery / gateway (aggregator)	0
Contribution	511	Contribution	339

An aggregator order earns **Rs.172 less contribution** (a 34% haircut) even after the aggregator absorbs delivery and payment. At 40% aggregator mix, blended CM falls to Rs.442 and break-even rises to ~15/day.

TWO CLARIFICATIONS ON THE COMMISSION (RAISED IN REVIEW)

(1) What is the 25% charged on? This model applies commission to the **ex-GST order value** (Rs.847) — the standard basis, since aggregators levy commission on the net menu value the restaurant sets, not the GST-inclusive bill. If the team's contract uses a gross basis, commission rises to ~Rs.250 (25% of ~Rs.1,000) and aggregator CM falls a further ~Rs.38; it's a one-line change in the model.

(2) Aggregators add 18% GST on their commission. That makes the *gross* commission $25\% \times 1.18 = 29.5\%$ (~Rs.250/order). But SliceMatic is GST-registered and **reclaims that Rs.38 as Input Tax Credit**, so the P&L cost of commission stays ~Rs.212 (25%) and the Rs.339 aggregator CM holds. The GST-on-commission is a **cash-timing** item (pay now, recover via ITC) — real for the cash-flow model, neutral for steady-state margin. That's why we keep it out of the CM but flag it here.

Note for the team: for orders placed *through* an aggregator, GST on the food itself is handled by the platform under Sec 9(5) of the CGST Act — a separate mechanism from the 18% we charge on our own direct deliveries. It doesn't alter the restaurant's contribution maths above, so it's out of scope for this model, but worth knowing if a grader probes GST.

STRATEGY: AGGREGATORS FOR ACQUISITION, OWNED CHANNEL FOR RETENTION

Treat the 25% commission as a **customer-acquisition cost**, not a delivery fee. It is worth paying *once* to win a customer — and a disaster to pay *forever*. The job of the ordering system is to capture that customer's phone number on first order and pull every subsequent order onto the owned channel, where the margin is Rs.172/order richer.

08 Customer Economics — CAC, LTV & Retention

Profitability is decided less by a single order than by **how many times a customer comes back**. Illustrative model (assumptions stated):

Metric	Value	Basis / assumption
Contribution / owned order	Rs.511	From Section 04

CAC — owned (organic/Meta)	~Rs.80	Rs.8k marketing ÷ ~100 new customers/mo
CAC — aggregator	~Rs.212	Commission on first order (recurring if not converted)
LTV — one-time buyer (4 orders)	Rs.2,044	Low-retention cohort
LTV — loyal buyer (18 orders/yr)	Rs.9,198	~1.5 orders/month for a year

OWNED CHANNEL — LTV:CAC ≈ 25:1

Rs.2,044 lifetime contribution against ~Rs.80 acquisition cost. Exceptional — *if* you can drive repeat behaviour.

AGGREGATOR-ONLY — LTV:CAC CAN BE < 1:1

If you pay Rs.212 commission on *every* order and never convert the customer, there is no payback — you are renting demand at a structural loss of margin.

This is why the Stage-3 recommendation engine is a business feature, not a gimmick: a 10% lift in repeat rate compounds directly into LTV and is the cheapest growth lever available.

09 Cash Flow & Ramp-Up

Operating profit and cash are different animals. The outlet can be **EBITDA-positive early but cash-negative for 18 months** because of the upfront Rs.15L investment.

Initial investment (Rs.15L)	Amount
Equipment (on EMI, Rs.4.5L financed)	Rs.4.5L
Fit-out	Rs.4.0L
Working capital	Rs.2.5L
Contingency + launch marketing	Rs.4.0L

- **Ramp curve:** 20% capacity in Month 1 climbing to ~60% by Month 6. Operating break-even (monthly) is reached around **Month 4-5**; cumulative cash recovers the Rs.15L by **~Month 18**.
- **Favourable working-capital cycle:** revenue is collected same-day (prepaid digital / COD), while ingredient suppliers often extend short credit. The business can run on **near-zero or negative working capital** once stable — a real cash advantage of QSR.
- **Burn risk is the fixed nut:** Rs.2.03L/month must be covered through the ramp. The danger window is Months 1-3, where low volume + aggregator drag keep EBITDA thin. Keep the Rs.4L contingency genuinely in reserve.

REFERENCE INCONSISTENCY

The reference's 58.9% operating margin assumes instant 60% capacity. If that were true, payback would be ~2-3 months, not 18. The 18-month payback is consistent only with the slow ramp — so the headline margin and the payback period describe two different worlds. Quote the ramp, not the steady-state snapshot, when describing launch.

10 Menu Engineering

Classic menu-engineering maps every item on two axes — **margin** and **popularity** — to decide what to push, fix, or cut.

Quadrant	Definition	SliceMatic example	Action
Stars	High margin + high volume	Margherita, Farm House	Feature prominently, protect quality
Plowhorses	Low margin + high volume	Cheese Burst (51% GM)	Re-engineer cost or nudge price up
Puzzles	High margin + low volume	Premium/specialty pizzas	Promote via recommendation engine
Dogs	Low margin + low volume	Slow non-sellers	Cut or rework

The order database makes this matrix *computable in real time* rather than guessed — directly linking the software to a margin decision the owner makes weekly.

11 Pricing & Discount Strategy

Sample bill (Stage-2 reference): $5 \times (\text{Cheese Burst Rs.229} + \text{BBQ Chicken Rs.379} + \text{Extra Cheese Rs.69}) = \text{Rs.3,385} \rightarrow -10\% \text{ discount (Rs.338.50)} \rightarrow \text{Rs.3,046.50} \rightarrow +18\% \text{ GST (Rs.548.37)} \rightarrow \text{Rs.3,594.87 payable.}$

Is the qty ≥ 5 / 10% discount a value-driver or a margin leak?

The 10% comes **straight out of contribution** — variable costs don't fall — cutting a 5-pizza order's CM by ~14%. But one large order replaces five single orders, saving ~4 delivery trips and gateway fees, which partly offsets the giveaway.

VERDICT & FIX

It is a **value-driver only if it changes behaviour** (pulls 3–4 pizza baskets up to 5) and a **pure leak** if customers who'd buy 5 anyway just pocket the discount. **Fix:** nudge at the threshold — show "add 1 more pizza, save 10%" at quantity 4 — so the discount captures incremental baskets rather than discounting inframarginal ones.

12 Delivery Radius Economics

Radius	Households	Delivery time	Trips/rider/day	Verdict
0–2 km	~18,000	8–12 min	55–60	Premium SLA
2–4 km	~55,000	15–22 min	30–35	Sweet spot
4–6 km	~1,10,000	25–40 min	18–22	Marginal — needs 3rd rider
> 6 km	—	40+ min	12–15	Not viable at launch

Launch at 4 km. The 3rd-rider decision is operational, not financial: a 3rd rider (~Rs.16k/month) pays for itself with under 1 extra order/day, but two riders saturate at ~60–70 deliveries/day. **Hire the 3rd rider when sustained volume nears ~55 orders/day** — to protect the SLA, not to cover its cost.

13 GST Treatment — Who Absorbs It?

The customer absorbs output GST; SliceMatic never books it as revenue. 18% GST on restaurant home-delivery is collected from the customer, held as a liability, and remitted monthly. The P&L is always computed ex-GST.

GST flow (monthly, at 60% cap)	Amount
Output GST collected from customers	~Rs.1,77,564
Less: Input Tax Credit (ingredients, packaging, equipment)	~(Rs.20,000)
Net GST payable to government	~Rs.1,57,564

Implementation rule (feeds the build): store menu prices **exclusive** of GST; add 18% on the **post-discount** subtotal at the billing step. Cash note: GST is collected before it is remitted, so it briefly improves cash — but it is not yours; ring-fence it.

14 KPIs & the Data-to-Decision Loop

Three business-intelligence metrics the order database unlocks — each tied to a specific profit lever the owner can act on:

Metric	Profit lever	Decision it drives
Peak-hour heatmap (orders × hour × weekday)	Labour & throughput	Staffing, prep batching, 3rd-rider timing — cut idle labour, recover lost peak orders
Basket / topping attach-rate & AOV drivers	AOV	Menu engineering & targeted upsell — lift the Rs.847 lever
Repeat-rate / RFM by phone number	Retention & commission	Convert aggregator customers to owned channel; protect CM

Supporting dashboard cadence — **Daily:** orders, AOV, CM, peak hour, SLA-hit %, owned-vs-aggregator mix. **Weekly:** repeat rate, top items, discount uptake, attach rate. **Monthly:** EBITDA, break-even coverage, CAC, cohort retention.

15 Risk & Assumptions Register

Risk	Severity	Mitigation
Demand cold-start (ramp slower than plan)	High	Aggregator-led launch for discovery; aggressive owned-channel conversion
Aggregator commission compression	High	Repeat loop + recommendation engine to shift mix to owned
Ingredient inflation (cheese is volatile)	Medium	Menu re-engineering, modest price increases (~Rs.36/order absorbs 12%)
Single-chef key-person risk	Medium	Cross-train helper; document recipes/SOPs
Rating / review shock	Medium	SLA discipline, deterministic billing (no pricing errors)
Rider attrition / fuel cost	Low-Med	Incentive structure; radius discipline at 4 km
Rent escalation	Low	Not the binding constraint (Section 05)

Key assumptions: AOV Rs.847 ≈ 2 pizzas/order · per-pizza ingredient Rs.130 · gateway 1.8% · aggregator commission 25% · mature capacity 60% (47 orders/day) · max kitchen throughput 80 orders/day. Change any of these and the model shifts — they are stated so a reviewer can challenge them.

16 Inconsistencies in the Reference Model

Flagging these is what separates analysis from transcription:

- **CM overstated ~26%** — per-pizza costs charged against per-order (multi-pizza) revenue. (Section 04.)
- **Page-1 "Monthly Revenue Rs.2.1L" is mislabelled** — Rs.2.03L is total *fixed cost*. Actual 60% revenue is Rs.11.73L gross / Rs.9.96L net.
- **Section 5 margin contradicts the Section 8 ramp** — 58.9% steady-state margin can't coexist with an 18-month payback (which implies a slow ramp).
- **Delivery cost double-count risk** — a per-order cost sits inside the per-pizza COGS table.

- **AOV basis ambiguity** — whether Rs.847 is GST-inclusive or ex-GST is never stated; this analysis treats it as ex-GST revenue per order and reconciles to ~2 pizzas.

17 Answers to the Stage-1 Challenge Questions

Q1 · Break-even if rent rises to Rs.70,000? At what rent is the model unviable?

Fixed cost → Rs.2,17,910; break-even rises from ~13 to **~14 orders/day** (corrected CM) — each Rs.10k of rent adds only ~0.6 orders/day. *In isolation*, the model only turns loss-making at planned 60% volume once total fixed costs approach monthly contribution (~Rs.7.2L), implying rent near ~Rs.5.7L — clearly unrealistic. But that single-variable figure is illustrative only: a larger/costlier space comes bundled with added staff and utilities, so the real test is the whole fixed-cost **block** against contribution. **Rent per rupee is low-sensitivity; demand and channel mix are the levers that decide viability.**

Q2 · 40% of orders via aggregator at 25% commission — new CM and break-even?

Aggregator order CM = Rs.339 (loses Rs.212 commission on ex-GST value, but saves own delivery + gateway). Blended CM = $0.6 \times 511 + 0.4 \times 339 = \text{Rs.442}$; new break-even = **~15 orders/day** (459/month). The 18% GST aggregators add on commission is reclaimed via ITC, so it doesn't change this P&L figure (see Section 07). Strategy: aggregators for acquisition, owned channel for retention.

Q3 · At what volume is a 3rd rider justified?

Financially: almost immediately — $\text{Rs.16k} \div \text{Rs.511 CM} \approx <1$ extra order/day pays for it. **Operationally:** two riders saturate at ~60-70 deliveries/day, so the binding trigger is **~55 orders/day**. Hire to protect the SLA, not to cover cost.

Q4 · Impact of the qty ≥ 5 / 10% discount — value-driver or margin leak?

Cuts a 5-pizza order's CM by ~14% (the discount comes straight from contribution). Value-driver if it lifts basket size, leak if inframarginal. Fix: a threshold nudge at quantity 4 ("add 1 more, save 10%") so it captures incremental orders only.

Q5 · COGS +12% — how many extra daily orders to hold EBITDA?

Ingredient cost +12% ≈ +Rs.36/order; CM falls Rs.511 → ~Rs.480. To hold the same monthly EBITDA you need **~+3 orders/day** (50/day vs 47), or a ~Rs.36/order price rise. Manageable.

Q6 · Three BI metrics from the order DB that improve profitability?

(1) Peak-hour heatmap → staffing & rider timing (labour lever). **(2)** Topping attach-rate & AOV drivers → upsell (AOV lever). **(3)** Repeat-rate / RFM by phone → owned-channel conversion (retention & commission lever). Each maps to a distinct profit driver.

BOTTOM LINE

SliceMatic is a **fundamentally healthy single-outlet QSR** with ~60% contribution margins and a ~13/day break-even it clears comfortably at target volume. The reference model is directionally right but optimistic by ~26% on margin. The business's fate rests on two levers the digital ordering system is purpose-built to move: **order volume** and **owned-channel mix**. Build the system to protect both, and the economics take care of themselves.